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Quant Training Session 6

Understanding Variance and Standard Deviation

Overview of Topics

- Variance
- Limiting Theorem

Variance

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Variance of a Random Variable

The variance of a random variable X, denoted $\mathrm{Var}(X)$, is calculated as:

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}X)^2\right] = \mathbb{E}X^2 - (\mathbb{E}X)^2$$

This measures the spread of X's values, indicating how much they deviate from the mean.

- Expanding $\mathbb{E}\left[(X - \mathbb{E}X)^2\right]$ gives us:

  $$\mathbb{E}\left[X^2 - 2\mu X + \mu^2\right] = \mathbb{E}X^2 - 2\mu\mathbb{E}X + \mu^2 = \mathbb{E}X^2 - \mu^2$$

  where $\mu = \mathbb{E}X$.
- For any constants a and b, we have:

  $$\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$$

Why square the differences? Squaring ensures differentiability and simplifies the expression for independent variables, such as in $\mathrm{Var}(X + Y) = \mathrm{Var}\,X + \mathrm{Var}\,Y$. And squaring is differentiable, making $\min \mathrm{Var}(Y - (aX + b))$ easy.

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Standard Deviation

The standard deviation of X, denoted as $\mathrm{SD}(X)$, is the square root of its variance:

$$\mathrm{SD}(X) = \sqrt{\operatorname{Var} X}$$

- Understanding units: If X is in dollars, $\operatorname{Var} X$ is in square dollars. We use the square root to return to the original units.
- For instance, the variance of a dice roll is $\operatorname{Var}(\text{ die roll }) = \left(2 \times 0.5^2 + 2 \times 1.5^2 + 2 \times 2.5^2\right)/6 = 35/12$ and $\mathrm{SD}(\text{ die roll }) = \sqrt{35/12} \approx 1.7$.
- Standardizing a random variable: To standardize Y with non-zero, finite variance, apply the transformation:

$$Z := \frac{Y - \mathbb{E}Y}{\mathrm{SD}(Y)}$$

This results in Z having a mean of 0 and a variance of 1:

$$\mathbb{E}Z = \mathbb{E}(Y - \mathbb{E}Y)/\mathrm{SD}(Y) = 0 \text{ and } \operatorname{Var} Z = \operatorname{Var} Y/\mathrm{SD}(Y)^2 = 1$$

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Interview Question

Question

Find the variance of the number of tosses required to get an "one" when using an n-sided die. Assume that each side of the die has an equal probability of landing.

Solution

Solution

Let's denote the number of tosses to get 1 as X. Since each toss is independent, X follows a geometric distribution with success probability $p = \frac{1}{n}$.

For a geometric distribution:

- The expected value (mean) $\mathbb{E}[X]$ is $\frac{1}{p} = n$.
- The variance $\mathrm{Var}(X)$ is given by $\frac{1-p}{p^2} = \frac{n-1}{n^2}$.

Therefore, the variance of the number of tosses to get a head for an n-sided die is $\boxed{\dfrac{n-1}{n^2}}$.

Covariance of Random Variables

Definition

The covariance of two random variables X and Y is defined as:

$$\mathrm{Cov}(X, Y) = \mathbb{E}((X - \mathbb{E}X)(Y - \mathbb{E}Y)) = \mathbb{E}(XY) - (\mathbb{E}X)(\mathbb{E}Y)$$

- Covariance measures if X and Y tend to increase or decrease together. A positive covariance indicates that X and Y generally move in the same direction, while a negative covariance means they move in opposite directions.

Equivalent Expressions

$$\begin{aligned}\mathbb{E}\left[(X - \mu_X)(Y - \mu_Y)\right] &= \mathbb{E}\left(XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y\right) \\ &= \mathbb{E}(XY) - \mu_X \mu_Y\end{aligned}$$

where $\mu_X := \mathbb{E}X$ and $\mu_Y := \mathbb{E}Y$.

- An interesting property: $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$.

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Understanding Correlation

Definition

For random variables X and Y with nonzero finite variances, their correlation, denoted as $\mathrm{Corr}(X, Y)$, is defined as:

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\mathrm{SD}(X)\,\mathrm{SD}(Y)}$$

- Correlation measures the strength and direction of a linear relationship between two variables.
- The sign of $\mathrm{Corr}(X, Y)$ (positive, negative, or zero) aligns with that of $\mathrm{Cov}(X, Y)$.

Properties and Examples of Correlation

- The correlation coefficient ranges between -1 and 1, inclusive: $-1 \leq \mathrm{Corr}(X, Y) \leq 1$.
- A correlation of 1 implies perfect positive linear correlation. For example, if $Y = aX + b$ with $a > 0$, then X and Y are perfectly positively correlated.
- Zero correlation or covariance indicates that X and Y are uncorrelated. However, uncorrelatedness does not necessarily imply independence.

Example

Consider two variables: the number of hours studied (X) and the score on a test (Y). If there is a consistent, predictable increase in Y with an increase in X, they are positively correlated. However, if studying more hours doesn't consistently affect the score, they might be uncorrelated.

Interview Question: Correlation and Independence Question

Question 1: Uncorrelated but Not Independent

Provide an example of two random variables that are uncorrelated but not independent. Explain why they are uncorrelated and how their dependence is exhibited.

Question 2: Bounding Correlation

Prove that the correlation coefficient, $\mathrm{Corr}(X, Y)$, for any two random variables X and Y, is always bounded between -1 and 1. Hint: Consider the Cauchy-Schwarz inequality and its implications for covariance and standard deviation.

Solutions: Correlation and Independence Question

Independence != Uncorrelated

Consider two random variables X and Y where X follows a normal distribution and $Y = X^2$. While X and Y are uncorrelated, they are not independent since knowing X gives information about Y.

Bounds of Covariance and Correlation

Given two random variables X and Y, the covariance and correlation are bounded by the product of their standard deviations:

$$|\,\mathrm{Cov}(X,Y)| \leq \mathrm{SD}(X)\,\mathrm{SD}(Y)$$

Consequently, this implies that:

$$-1 \leq \mathrm{Corr}(X,Y) \leq 1$$

This bound is derived using the Cauchy-Schwarz inequality.

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Bilinearity Property of Covariance

General Form

The covariance of linear combinations of random variables is given by:

$$\operatorname{Cov}\left(\sum_j a_j X_j, \sum_k b_k Y_k\right) = \sum_j \sum_k a_j b_k \operatorname{Cov}\left(X_j, Y_k\right)$$

Special Case: Variance of a Sum

Considering the sum of two random variables X and Y:

$$\begin{aligned}
\operatorname{Var}(X+Y) &= \operatorname{Cov}(X+Y, X+Y) \\
&= \operatorname{Cov}(X,X) + \operatorname{Cov}(X,Y) + \operatorname{Cov}(Y,X) + \operatorname{Cov}(Y,Y) \\
&= \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y)
\end{aligned}$$

This leads to an important result: if X and Y are uncorrelated, then $\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y)$.

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Variance of a Sum of Random Variables

General Formula

For random variables $X_1, X_2, \ldots, X_n$:

$$\mathrm{Var}\left(X_1 + \cdots + X_n\right) = \sum_{j} \mathrm{Var}\left(X_j\right) + 2\sum_{j<k} \mathrm{Cov}\left(X_j, X_k\right)$$

- This formula shows how the variance of a sum includes not just the variances of individual components but also their pairwise covariances.
- It highlights the importance of understanding how variables interact with each other in terms of their covariances.

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Expectation and Portfolio Implications

Comparison with Expectation

In contrast to variance, the expectation of their sum is simpler:

$$\mathbb{E}\left(X_1 + X_2 + \cdots + X_n\right) = \mathbb{E}\left(X_1\right) + \mathbb{E}\left(X_2\right) + \cdots + \mathbb{E}\left(X_n\right)$$

- Unlike variance, the expectation of the sum is unaffected by the correlations between the variables.
- In portfolio theory, while the expected return of a portfolio is the sum of the expected returns of its assets, the portfolio's risk (variance) also depends on how the assets interact (covariance).

Limiting Theorem

Chebyshev's Inequality - Statement

Formulation of the Inequality

Given a random variable X with a finite variance, for any positive constant c, Chebyshev's inequality is stated as:

$$\mathbb{P}(|X - \mathbb{E}X| > c) = \mathbb{E}\left(\mathbf{1}_{|X-\mathbb{E}X|>c}\right) \leq \frac{\operatorname{Var} X}{c^2}$$

- This inequality bounds the probability that X deviates from its mean $\mathbb{E}X$ by more than c.
- It's a measure of dispersion and is particularly powerful because it does not depend on the distribution of X, only on its variance.

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Chebyshev's Inequality - Application and Significance

Practical Application

Setting $c = n \cdot \mathrm{SD}(X)$, where n is a positive number, demonstrates that the probability of X differing from its mean by more than n standard deviations is at most $\frac{1}{n^2}$.

- This aspect of Chebyshev's inequality is crucial for understanding how extreme deviations from the mean are unlikely, particularly for large values of n.
- It is a fundamental tool in statistics for assessing the "spread" or "variability" of a data set, especially in contexts where little is known about the distribution's shape.

Weak Law of Large Numbers - Introduction

Setting and Basic Equations

Consider a sequence of independent and identically distributed random variables $X_1, X_2, X_3, \ldots$, each with:

$$\mu = \mathbb{E}X \text{ and } \sigma = \mathrm{SD}(X) < \infty$$

Define their sum as $S_n := X_1 + X_2 + \cdots + X_n$, which gives:

$$\mathbb{E}S_n = \mu n \text{ and } \operatorname{Var} S_n = n\sigma^2$$

Sample Mean

The sample mean $\bar{X}_n$ is given by $\bar{X}_n := S_n/n$, leading to:

$$\mathbb{E}\bar{X}_n = \mu \text{ and } \operatorname{Var} \bar{X}_n = \frac{\sigma^2}{n}$$

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Weak Law of Large Numbers - Conclusion

Convergence in Probability

By applying Chebyshev's inequality, we find that for any $\varepsilon > 0$:

$$\mathbb{P}\left(\left|\bar{X}_n - \mu\right| > \varepsilon\right) \leq \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0 \text{ as } n \to \infty$$

This implies $\bar{X}_n \to \mu$ in probability, which is the essence of the Weak Law of Large Numbers (WLLN).

- WLLN states that as the sample size n grows, the sample mean $\bar{X}_n$ gets closer to the population mean μ in probability.
- This law underpins many statistical methods, emphasizing the stabilizing effect of large sample sizes on sample means.

Strong Law of Large Numbers (SLLN)

Introduction to SLLN

The Strong Law of Large Numbers (SLLN) is an extension of the Weak Law of Large Numbers (WLLN), with a stronger form of convergence:

- It assumes only $\mathbb{E}|X| < \infty$.
- It guarantees that the sample mean $\bar{X}_n$ converges to the population mean μ "almost surely":

$$\mathbb{P}\left(\bar{X}_n \to \mu\right) = 1$$

- "Almost sure" convergence is a stronger condition than convergence in probability, as ensured by the WLLN.
- It implies that the sample mean will converge to the population mean for almost all sample paths.

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Central Limit Theorem (CLT)

Introduction to CLT

The Central Limit Theorem (CLT) provides another direction for extending the WLLN, focusing on the distribution of the sample mean for large sample sizes:

- It states that for large n, the distribution of $\bar{X}_n$ is approximately normal.

$$\mathbb{P}\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq x\right) \longrightarrow \Phi(x) \text{ as } n \to \infty$$

where Φ is the CDF of the standard Normal distribution.

- This theorem is foundational in statistics, underpinning many inferential techniques.
- It highlights the normality of sample means, regardless of the original distribution, given a sufficiently large sample size.

Interview Question: Coin Toss Probabilities

Question

Consider a fair coin being tossed multiple times:

1. If the coin is tossed 10 times, what is the probability of getting exactly 5 heads?
2. What is the probability of getting exactly 50 heads when the coin is tossed 100 times?
3. Extend this to 10,000 coin tosses. What is the probability of getting exactly 5,000 heads?

- This question explores the concept of binomial probabilities and the application of the Central Limit Theorem in different scenarios.
- Think about how the probability distribution changes as the number of coin tosses increases.

Solution: Applying the Central Limit Theorem

Binomial Distribution and CLT

The number of heads in coin tosses follows a binomial distribution, $B(n, p)$, where n is the number of tosses and $p = 0.5$ is the probability of getting a head. For large n, this distribution can be approximated using a normal distribution, as per the CLT.

Central Limit Theorem

The CLT states that if $X_1, X_2, \ldots, X_n$ are i.i.d. random variables with mean μ and variance σ^2, then the distribution of the sample mean $\bar{X}_n$ approaches a normal distribution with mean μ and variance σ^2/n.

- In our case, each coin toss is a Bernoulli trial with $\mu = p = 0.5$ and $\sigma^2 = p(1 - p) = 0.25$.
- For large n, the distribution of the number of heads is approximately normal with mean $np = n/2$ and variance $np(1 - p) = n/4$.

Solution: Approximation Steps

Approximation for Large n

To approximate the probability of getting exactly k heads in n tosses:

1. Normalize the binomial random variable: $Z = \frac{X-np}{\sqrt{np(1-p)}}$.
2. For $n = 100$ and $k = 50$, calculate $Z = \frac{50-50}{\sqrt{25}} = 0$.
3. Use the standard normal distribution to find $P(Z \leq z)$ for the calculated z value.
4. Adjust for the fact that we need $P(Z = z)$, which can be approximated by $P(z - 0.5 < Z < z + 0.5)$.

- This method provides a close approximation for the probability of getting exactly 50 heads in 100 tosses, and similarly for 5,000 heads in 10,000 tosses.
- As n grows, the approximation becomes more accurate due to the CLT.

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Interview Question: Estimating Minimum Sample Size

Question

A researcher wants to estimate the probability of monkeys having a disease with a high degree of certainty. She collects data and calculates an estimate $\widehat{p}$ for the true probability p. What is the minimum sample size required to ensure with at least 99% certainty that the difference between $\widehat{p}$ and p is no more than 10%?

- This question tests understanding of sample size calculations, statistical confidence, and margins of error.
- Consider how these concepts are interconnected and used in practical research scenarios.

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Solution: Estimating Minimum Sample Size

Solution Approach

The solution is based on the concept of confidence intervals and z-scores. For a 99% confidence level, the z-score is approximately 2.58.

Calculation

Given the margin of error $E = 10\% = 0.10$ and the estimate $\widehat{p}$, the equation for the sample size n is:

$$2.58\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} = 0.10$$

Solving for n, we get:

$$n = \frac{2.58^2 \times \widehat{p}(1-\widehat{p})}{0.10^2}$$

Note: $\widehat{p}$ must be provided or estimated for a specific calculation.

Solution: Estimating Minimum Sample Size

- This calculation provides the minimum sample size needed to achieve the desired confidence level and margin of error.
- It is a key concept in designing statistically sound research studies.

Interview Question: CLT for Medians or Quantiles

Question

In real-world data analysis, especially with data containing anomalies, we often rely on medians or quantiles instead of means. Can the Central Limit Theorem be applied to the medians or fixed quantiles of sample data? If so, how can it be done, and what are the implications for statistical analysis?

- Consider how the CLT, typically applied to means, might extend to other measures like medians or quantiles.
- Explore the potential challenges and requirements for this application.

Solution: CLT for Medians or Quantiles (Part 1)

Applying the CLT to Indicator Variables

By expressing the problem in terms of indicator variables (e.g., $Z_i = 1$ if $X_i \leq x$ and 0 otherwise), we can apply the CLT to the mean of Z's. This application leads to an asymptotic normal distribution for $F_X^{-1}(\bar{Z})$, where F_X^{-1} is the inverse CDF.

Implication for Fixed Quantiles

The approach implies that not only the median but also other quantiles (like quartiles, 90th percentiles, etc.) will have an asymptotic normal distribution in large samples.

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Solution: CLT for Medians or Quantiles (Part 2)

Approximation for Quantiles

For the qth sample quantile, in large samples, we have an approximate normal distribution with mean x_q (the qth population quantile) and variance $\frac{q(1-q)}{nf_X(x_q)^2}$.

Specific Case: Median

For the median ($q = 1/2$), the variance in large samples is approximately $\frac{1}{4nf_X(\tilde{\mu})^2}$, where $\tilde{\mu}$ is the population median.

- Conditions like continuity, positive density at the quantile, and differentiability are required for this approximation to hold.
- It does not apply to extreme quantiles, where different theoretical considerations are necessary.

Bonus: Clarifying the Central Limit Theorem

Avoiding Misinterpretation

A common misconception is to equate the normal distribution of sample means (as per the CLT) with the distribution of the data itself. The CLT's application to means does not justify assuming normality in the underlying data distribution.

Practical Considerations

The utility of the CLT lies in comparing sample means from non-normally distributed data. However, one must first ascertain whether comparing means is appropriate given the data's distribution.

- The CLT applies under conditions of independent and identically distributed random variables with finite variance.
- The comparison or tests based on CLT should be meaningful and relevant to the research question and the nature of the data.

Anecdote: Misconceptions about the CLT in Practice

A Common Misunderstanding

A common yet incorrect assumption is that having a large dataset implies normality due to the CLT. For instance, it might seem that in a large dataset (say 50,000 observations), the data must be close to normal, allowing for the application of the empirical rule (e.g., 68% of data within one standard deviation).

Anecdote: Misconceptions about the CLT in Practice

It's wrong

This interpretation is a misconception. The CLT does not claim that the empirical distribution of the data converges to normality.

- The size of the dataset does not change the underlying distribution of the population.
- The empirical distribution of the data remains consistent with the population distribution, regardless of sample size.
- Emphasize that the CLT applies to the behavior of sample means, not to the dataset as a whole.

Intuition Behind the CLT - The Balancing Act

The Sum of Random Variables

Picture this: You're adding up a bunch of random numbers, say $X_1, X_2, \ldots, X_n$. Sometimes one number is a bit low, sometimes another is a bit high. It's like a balancing act. The low numbers get evened out by the high ones. This is the essence of why, when you add a lot of these numbers, their sum starts to behave in a predictable way.

- Imagine each number in the sum doing a little dance around its average. As they all dance together, their individual steps, sometimes big, sometimes small, start to smooth each other out.
- This 'smoothing out' is a key part of why the sum of many random variables has a kind of 'normal' behavior.

Intuition Behind the CLT - Why Normality?

From Dance to Normality

So, we've got this smooth dance of numbers. But why does this dance lead to a normal distribution? Well, that's a bit tricky. It comes down to some heavy-duty math involving moment-generating functions, a type of math formula that characterizes probability distributions.

- Think of it like this: There's a mathematical 'shape' that emerges from all these dances, and that shape looks a lot like a bell curve, the normal distribution.
- It's not so much about the steps of the dance, but about how these steps, when added up in a particular way, form a pattern that mathematicians recognize as 'normal'.

Intuition Behind the CLT - Limitations and Misconceptions

It's Not Really Normal

Here's the catch: No matter how many numbers you add, their sum doesn't actually become a perfect bell curve. It just starts to look more like one.

The Practical Side

In real life, when you use the CLT, it's great for understanding the average of your numbers. But be careful - it's not so great for making guesses about the extremes, the very high or very low numbers. That's where the CLT can lead you astray.

- Remember, the CLT is a fantastic tool, but it's not a magic wand. It helps with averages, not with the wild, unusual stuff.
- Next time someone says, "We have lots of data, so it must be normal," you'll know it's not that simple.

THE END